

**Information Technology University of the Punjab**  
**SE460T Cloud Computing with Big Data Applications – Fall 2025**  
**Quiz 3 – December 17, 2025**

Name: \_\_\_\_\_

Roll No.: \_\_\_\_\_

Time allowed: 25 minutes

Maximum Marks: 23

1. ETL and ELT are related to all stages of a data pipeline except which one? [1]

Analysis and Visualization

2. What are the 6 steps in data wrangling? Just provide the terms for the steps without any details. [3]

Discover, Clean, Structure, Enrich, Validate, Publish

3. Which AWS service is used to ingest data from SaaS apps? [1]

Amazon Appflow

4. Provide the name of an AWS service that can serve as a data lake. [1]

Amazon S3

5. An ELT pipeline typically loads the data in a data lake and an ETL pipeline loads in a data warehouse. [1]

6. In the Discovery phase of data wrangling, give an example that illustrates the meaning of identifying relationships between sources. [2]

Ticket table in customer support system (supp1)

| ticket_id | requestor_id | submitter | assignee | group | subject | status | priority | ticket_type | create_date | updated_date | solved_date |
|-----------|--------------|-----------|----------|-------|---------|--------|----------|-------------|-------------|--------------|-------------|
|           |              |           |          |       |         |        | High     |             |             |              |             |
|           |              |           |          |       |         |        | Low      |             |             |              |             |

Ticket table in the acquired customer support system (supp2)

| issue_id | cust_num | description | status | priority | create_date | updated_date | closed_date |
|----------|----------|-------------|--------|----------|-------------|--------------|-------------|
|          |          |             |        | 1        |             |              |             |
|          |          |             |        | 3        |             |              |             |

Relationship identification:

supp1 requestor\_id = supp2 cust\_num

supp1 ticket\_id = supp2 issue\_id

supp1 subject = supp2 description

7. What activities are performed during the data validation phase of data wrangling? [2]

Audit your work

Count expected rows. Check consistency. Check expected formats and data types. Check for duplicates. Check for PII. Check for outliers.

Fix the data

Address audit findings. Address issues at the source. Fix issues with integration tools or scripts.

8. What are the different ways in which you can author jobs in AWS Glue? [2]

Visual ETL

Notebook

Script

9. What is meant by orchestration in batch ingestion?

[2]

Orchestration in batch ingestion means coordinating and managing the sequence, dependencies, scheduling, and monitoring of batch data processing steps—ensuring each task (ingest, transform, load) runs in the correct order, at the right time, and handles failures properly.

10. What does the AWS DataSync service do? At what stage or stages of the data pipeline is it used?

[2]

AWS DataSync is used to ingest data from file systems. We can apply filters to transfer a subset of files using variety of file systems as sources and target.

It is used primarily at the ingestion stage of the data pipeline. The transformation that DataSync allows can be considered as part of the processing stage of the pipeline.

11. What is data cataloging? Which AWS Glue feature automatically performs cataloging on a given dataset? Where is the data catalog stored once Glue creates it?

[3]

Data cataloging is the process of discovering, classifying, and storing metadata about datasets.

Glue Crawler

In a database created inside Glue Catalog.

12. Which stage or stages of the data pipeline do AWS Glue Jobs belong to?

[1]

Transformation (Processing)

13. How can you ensure that one job in an AWS Glue Workflow runs only when another has completed its run?

[2]

By adding a trigger at the end of Job A and attaching the trigger to Job B. In the trigger, the option selected should be of “event”.